



NAUVAL ZULFIKAR

AI Safety & Alignment Researcher · Trustworthy ML Candidate Bandung, Indonesia (relocatable to Tübingen, 2026-10-01) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264
[LinkedIn](#) · [GitHub](#) · [Portfolio](#) **Visa status:** UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

AI Safety & Alignment Researcher with a strong foundation in ML and NLP, demonstrated through applied projects in licensed banking, government, and start-up environments. **MSc Business Analytics, Aston University, 1:1 First Class.** Dissertation focused on blockchain-integrated LLM analysis for supply-chain systems, supervised by Dr. Ammar Al-Bazi. Passionate about advancing AI safety and trustworthy ML, aligning with Dr. Maksym Andriushchenko's research themes.

EDUCATION

Aston University, Birmingham, UK (*MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024*)

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan (*BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019*)

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- Distinctions: Business Data Analysis, Consumer Behaviour, Marketing Research

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules (2026)

- Developed an end-to-end RAG system using gpt-4o-mini, focusing on reproducibility and ablation studies.
- [github.com/nauvalZulfikar/LLM-Shipper-Profiles](#)

Crowd-Shipping ABM Behavioural Sweep (2026)

- Conducted a 450-run parameter sweep to evaluate behavioural modelling, achieving significant insights into supply effects.
- [github.com/nauvalZulfikar/Crowd-Shipping-ABM-](#)

VRPOD MILP Optimisation (2026)

- Implemented MILP optimisation for VRPOD, achieving a 38.5% cost reduction on a Birmingham-inspired instance.
- [github.com/nauvalZulfikar/VRP-MILP](#)

Mini Digital Twin for Integrated Passenger-Freight Transport (2026)

- Built a three-layer Digital Twin prototype with interactive dashboards and live KPIs.
- [github.com/nauvalZulfikar/Mini-Digital-Twin](#)

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung (*Data Scientist & Project Manager · Sep 2022 – Present*)

- Automated ETL pipelines and dashboards, enhancing project delivery by 45% and data efficiency by 20%.

Syncwell, Birmingham, UK (*Marketing Data Scientist · Jul – Sep 2024*)

- Implemented predictive ML strategies, increasing CTR by 40% and follower growth by 22%.

PCOS Challenge, Birmingham, UK (*Strategic Data Scientist Specialist · Jun – Aug 2024*)

- Developed models and visualisations, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta (*Lead Data Scientist · Nov 2019 – Aug 2022*)

- Created NLP models for risk scoring and market analysis, achieving a 5.3% investment return.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Working paper evaluating ML regressors for retail sales forecasting.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API, RAG, sentence-transformers, classification, regression, time-series forecasting, recommender systems, attribution modelling.

Languages/Infrastructure: Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, FastAPI, Streamlit, AWS, Docker, Git, ETL/orchestration.

Mathematical/Formal: MILP (Pyomo, scipy.optimize.milp, HiGHS), Agent-Based Modelling (mesa), Discrete-Event Simulation (simpy), Digital Twin architecture, Bayesian methods, causal inference.

Governance: Model validation, reliability, audit-grade documentation, model staleness monitoring, A/B testing, Power BI, Tableau.

LANGUAGES

Indonesian (*Native*) · **English** (*Fluent, IELTS 7.0 overall: L7.5/R7.5/W6.5/S7.0*) · **Japanese** (*Intermediate*)

REFEREES

Dr. Ammar Al-Bazi

Senior Lecturer, Aston Business School, Aston University

Email: a.al-bazi@aston.ac.uk

Relationship: MSc dissertation supervisor

Dr. Viktor Pekar

Lecturer, Aston Business & Social Sciences

Email: v.pekar@aston.ac.uk

Relationship: MSc Personal Tutor; taught Machine Learning for Business Analytics + Research Methods